PROJECT REPORT

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PROJECT : HOTEL MANAGEMENT SYSTEM

COURSE : FUNDAMENTALS IN AI AND ML

COURSE CODE : CSA2001

INTRODUCTION

- **THIS PROJECT IS A BASIC HOTEL MANAGEMENT SYSTEM MADE USING PYTHON.**
- **THE PROGRAM HELPS IN MANAGING HOTEL ROOMS BY ALLOWING CHECKIN,CHECKOUT & SHOWING WHICH ROOMS ARE VACANT OR OCCUPIED.**
- **THE AIM OF THIS PROJECT IS TO LEARN AND UNDERSTAND FILE HANDLING,LOOPS,FUNCTIONS IN PYTHON WHILE BUILDING SOMETHING THAT CAN BE USED IN REAL LIFE.**
- **THIS PROGRAM CAN STORE ROOM DETAILS AMD TRACK GUEST CHECKIN & CHECKOUT DATE EASILY.**
- **THE PROGRAM MAKES THE HOTEL WORKFLOW EASIER BECAUSE THE USER CAN QUICKLY CHECK WHICH ROOMS ARE VACANT AND WHICH ROOMS ARE ALREADY OCCUPIED**

ALGORITHM

1. START
2. LOAD ROOM DATA FROM THE FILE OR CREATE NEW FILE IF NO FILE EXISTS.
3. DISPLAY MENU AND ASK THE USER TO CHOOSE AN OPTION
4. BASED ON THE CHOICE , PERFORM THE SELECTED ACTION :
    a) CHECK-IN
    b) CHECK-OUT
    c) SHOW ROOMS(VACANT/OCCUPIED/ALL)
5. SAVE THE UPDATED DATA TO THE FILE
6. REPEAT THE MENU UNTIL USER SELECTS EXIT
7. STOP

PROGRAM

INPUT :

```python
import json
import os
from datetime import datetime

datafile= "roomsdata.json"

def loadrooms():
    if os.path.exists(datafile):
        with open(datafile) as file:
            try:
                return json.load(file)
            except:
                return {}
    else:
        return {}

def saverooms(rooms):
    with open(datafile, "w") as file:
        json.dump(rooms, file, indent = 4)

rooms = loadrooms()

if not rooms :
    for floor in range(1,6):
        for room in range(1,11):
            roomnumber = str(floor) + str(room).zfill(2)
            rooms[roomnumber] = { "status" : "vacant",
                                  "guest": "",
                                  "checkin" : "",
                                  "checkout" : ""}

def checkin():
    print("==Check In==")
    room = input("Enter room no. :")
    if room not in rooms:
        print("Wrong room no.")
    elif rooms[room]["status"]== "occupied":
        print("This room is occupied already")
    else:
        guestname = input("Guest name :")
        checkindate=str(datetime.now().date())
        rooms[room]["status"] = "occupied"
        rooms[room]["guest"] = guestname
        rooms[room]["checkin"] = checkindate
        rooms[room]["checkout"] = ""
        saverooms(rooms)
        print("Checkin done !")

def checkout():
    print("==Check Out==")
    room = input("Enter room no. :")
    if room not in rooms:
        print("Wrong room no,")
        return
    if rooms[room]["status"] == "vacant":
        print(" The room is already empty")
    else:
        checkoutdate=str(datetime.now().date())
        rooms[room]["status"] = "vacant"
        rooms[room]["guest"] = ""
        rooms[room]["checkout"] = checkoutdate
        rooms[room]["checkin"] = ""
        saverooms(rooms)
        print("Checkout done !")

def show_all_rooms():
    print("==ALL ROOMS==")
    for roomnumber,info in rooms.items():
        print(f"Room {roomnumber} - {info['status']}")

def show_vacant_rooms():
```

```python
    print("== VACANT ROOMS ==")
    found = False
    for roomnumber,info in rooms.items():
        if info["status"].strip() == "vacant":
            print(roomnumber)
            found = True
    if not found:
        print("No Vacant Rooms")

def show_occupied_rooms():
    print("== OCCUPIED ROOMS ==")
    found = False
    for roomnumber,info in rooms.items():
        if info["status"] == "occupied":
            print(f"{roomnumber} - Guest:{info['guest']}")
            found = True
    if not found:
        print("No Occupied Rooms")

def main():
    while True:
        print("== HOTEL MANAGEMENT SYSTEM ==")
        print("1. Check In")
        print("2. Check Out")
        print("3. Show Vacant Rooms")
        print("4. Show Occupied Rooms")
        print("5. Show All Rooms")
        print("6. Exit")

        choose = input("Pick Option :")

        if choose == "1":
            checkin()
        elif choose == "2":
```

```python
            checkout()
        elif choose == "3":
            show_vacant_rooms()
        elif choose == "4":
            show_occupied_rooms()
        elif choose == "5":
            show_all_rooms()
        elif choose == "6":
            print("Thank You for using this service")
            break
        else:
            print("Wrong Input")

main()
```

OUTPUT:

```
== HOTEL MANAGEMENT SYSTEM ==
1. Check In
2. Check Out
3. Show Vacant Rooms
4. Show Occupied Rooms
5. Show All Rooms
6. Exit
Pick Option :
```

GITHUB REPOSITORY: https://github.com/paarth-chaurasia/hotelmanagement

CONCLUSION

- **THE PROJECT EXECUTED CORRECTLY AND THE HOTEL MANAGEMENT FUNCTIONS WORK AS INTENDED.**
- **THE PROGRAM SAVED AND UPDATED DATA PROPERLY, GIVING THE EXPECTED OUTPUT**

THANK YOU